CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

4th Semester of B. Tech. Examination (EE)

May-2014

EE-204/EE-204.01 ELCTRICAL POWER SYSTEM-I

Date: 05.05.2014, Monday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Define String efficiency. Can its value be 100%? Also derive the mathematical expression of string efficiency for 3-disc string. **[07]**

Q - 2 (a) A string of 4 insulators has self-capacitance equal to 4 times the pin-to-earth capacitance. Calculate (i) the voltage distribution across various units as a percentage of total voltage across the string and (ii) string efficiency. **[07]**

(b) What do you mean by safety factor of insulator? Explain the causes of insulator failure in case of pin type insulator. **[04]**

(c) Draw Single line diagram of AC power supply scheme and mention basic components of power system network. **[03]**

OR

Q - 2 (a) Define depreciation and explain diminishing value method for measurement of Depreciation. **[07]**

(b) Give objectives of tariff and explain 3-part tariff and block rate tariff in detail. **[07]**

Q - 3 (a) Why grading of cable is necessary? Explain Capacitance grading with neat sketch. **[07]**

(b) Define Most economical power factor. With necessary diagram Prove that it is independent of the original power factor. **[07]**

OR

Q - 3 (a) Define dielectric stress. Also derive the Ratio between maximum potential gradient to minimum potential gradient for single core cable. [07]

(b) Give the disadvantages of low power factor. [04]

(c) Derive the equation of Insulation resistance of a single core cable with necessary diagram. [03]

SECTION – II

Q - 4 (a) Give the classification of overhead transmission line. Also draw the equivalent circuit and phasor diagram of single phase short transmission line and derive the equation of % voltage regulation and transmission efficiency. [07]

Q - 5 (a) Derive the equation of Inductance of single phase two wire line. [07]

(b) A 3-phase, 50 Hz transmission line 100 km long delivers 20 MW at 0.9 p.f. lagging and at 110 kV. The resistance and reactance of the line per phase per km are 0.2 Ω and 0.4 Ω respectively, while capacitance admittance is $2.5*10^{-6}$ siemen/km/phase. Calculate: (i) the current and voltage at the sending end (ii) efficiency of transmission. **Use nominal T method.** [07]

OR

Q - 5 (a) Derive the equation of flux linkages due to a single current carrying conductor due to internal flux and external flux effect. [07]

(b) The generalized constants of one phase of a 3-phase line are (1) A=D= 0.9 +j 0.012 (2) B= (22.5 + j 150) Ω (3) C= (- 0.00004 + j 0.001) ℧. The sending end voltage is 240 kV and the receiving end voltage is 220 kV line to line. Draw the circle diagram and determine the active and reactive power received when the angle between sending end and receiving end voltage is 30^0. [07]

Q - 6 (a) Explain Nominal π method with phasor diagram for medium transmission line and derive the equation of sending end voltage and sending end current. [07]

(b) A 3-phase, 50 Hz, 132 kV overhead line has three conductors placed in a horizontal plane, adjacent distance between each conductor is 4 m. Conductor diameter is 2 cm. If the line length is 100 km, calculate the charging current per phase assuming complete transposition. **[05]**

(c) Define skin effect. On which factors skin effects depend? **[02]**

OR

Q - 6 (a) Derive the equation of capacitance of symmetrically spaced double circuit 3-phase line using GMD and GMR concept. **[08]**

(b) As shown in **Fig.1** the spacings of a double circuit 3-phase overhead line. The phase sequence is ABC and the line is completely transposed. The conductor radius is 1.3 cm. Find the inductance per phase per kilometre. **[06]**

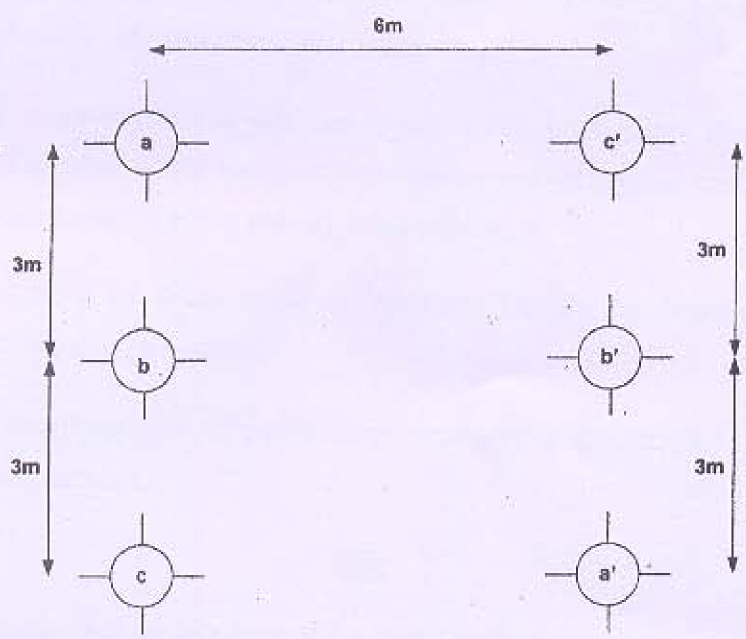

Fig.1

BEST OF LUCK

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